Siqi WANG

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Education

University of California, San Diego Sept. 2024 – Mar. 2026 (Expected)
Master of Science in Elecrical and Cumputer Engineering, GPA: 3.9/4.0

Shanghai Jiao Tong University Sept. 2020 – June 2024
Bachelor of Engineering in Information Engineering, GPA: 3.4/4.0

Experiences

Responsible Image Editing and Text-to-Image Generation Apr. 2025 – Present
Research Intern *SVCL Lab, UC San Diego*

- Built a responsible image editing pipeline that masks risky regions via GPT-based semantic segmentation, followed by Stable Diffusion inpainting conditioned on GPT-generated safe textual descriptions.
- Enhanced classifier-free guidance in diffusion models by applying orthogonal projection of negative prompts, reducing unsafe generation rate from 0.247 to 0.211 and validating gains via FID and CLIP scores.
- Developed and trained a negative embedding learning framework that maps unsafe prompts to safe latent representations, leveraging paired dataset supervision and latent-space MSE consistency.

Medical Image Classification via RGB Channel and Depth Information June 2023 – July 2024
Research Intern *Shanghai Jiao Tong University*

- Built a two-branch dermoscopy image classifier, extracting channel-specific features (via separable convolutions) and channel-shared features (via ResNet-50), using differential loss to separate representations.
- Designed multimodal medical imaging models by fusing polarized and non-polarized dermoscopy modalities with attention blocks, improving complementary feature integration and classification accuracy.
- Enhanced volume-based OCT classification by converting images into videos and applied Timesformer-based temporal modeling, capturing continuity across frames and improving accuracy by 15% over baselines.
- Presented a poster at IEEE International Symposium on Biomedical Imaging (ISBI) 2024.

Selected Projects

LLM-Based Agent for Automated Weather Disaster Management Mar. 2025 – June 2025

- Developed a hybrid LangGraph agent that integrates a predictive ML model with an LLM to assess weather risks, determine severity, and initiate automated responses and alerts.
- Trained and tuned several classical machine learning and deep learning models on historical climate datasets for wildfire prediction, selecting Random Forest as the optimal model for deployment.
- Developed a stateful agent workflow (via Gemini API and LangGraph) that synthesizes ML and LLM outputs to process real-time weather data, autonomously generating emergency response plans.

Zero-Shot Semantic Segmentation with Image Deraining Sept. 2024 – Dec. 2024

- Implemented a two-stage framework that integrates image restoration for deraining and zero-shot semantic segmentation without prior knowledge.
- Developed a guided filter-based deraining algorithm that decomposes rain images into low- and high-frequency parts, with edge enhancement and multi-stage refinement to produce derained images.
- Leveraged the diffusion UNet encoder to extract features from the last self-attention block, followed by recursive graph partitioning and high-resolution concept assignment for accurate segmentation.

Language-Queried Audio Source Separation (LASS) Jan. 2025 – Mar. 2025

- Designed a lightweight framework for LASS by integrating NMF with pretrained CLAP embeddings.
- Introduced a dynamic selection module to merge audio components based on semantic similarity to text queries.
- Delivered a training-free, interpretable solution for resource-constrained devices and real-time applications.

Publication

Yaoqi Tang, Qilin Sun, **Siqi Wang**, Bo Jiang, and Yuye Ling. Classifying Melanocytic Nevus by Using Extracted Depth-Encoded Channel Information. IEEE International Symposium on Biomedical Imaging (ISBI), 2024.

Skills

Programming Languages: Python, C/C++, R, Bash, MATLAB, HTML/CSS, JavaScript
Tools and Frameworks: PyTorch, TensorFlow, Keras, Hugging Face, OpenCV, Spark, Docker, Kubernetes, Git